1

(a) $P(\text{toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = \boxed{0.2}$

(b) $P(\text{Cavity}) = 0.108 + 0.012 + 0.072 + 0.008 = \boxed{0.2}$

(c) $P(\text{Toothache} \mid \text{Cavity}) = \dfrac{P(\text{Toothache} \wedge \text{Cavity})}{P(\text{Cavity})} = \dfrac{0.108 + 0.012}{0.2} = \boxed{0.6}$

(d) $P(\text{Cavity} \mid \text{toothache} \vee \text{catch}) \Rightarrow \dfrac{0.12 + 0.18 - 0.108}{0.2 + 0.34 - 0.124} \Rightarrow \dfrac{0.192}{0.416} = \boxed{0.4615} \approx \boxed{0.462}$

2 Given: $P(V) = 0.01$, $P(\neg V) = 0.99$

$P(A \mid V) = 0.95$, $P(A \mid \neg V) = 0.10$, $P(B \mid V) = 0.90$, $P(B \mid \neg V) = 0.05$

$$P(V \mid A) = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.10 \times 0.99} = \frac{0.0095}{0.1085} = \underline{0.0876}$$

$$P(V \mid B) = \frac{0.9 \times 0.01}{0.9 \times 0.01 + 0.05 \times 0.99} = \underline{0.1538}$$

$\boxed{\text{Test B is more indicative}}$ of really carrying the virus.

3. Given: T: test positive, D: has disease

$P(T \mid D) = 0.99$, $P(\neg T \mid \neg D) = 0.99$, $P(D) = 0.0001$

$$P(D \mid T) = \frac{P(T \mid D)P(D)}{P(T \mid D)P(D) + P(T \mid \neg D)P(\neg D)} \Rightarrow \frac{0.99 \times 0.0001}{0.99 \times 0.0001 + (1 - 0.99) \times (1 - 0.0001)}$$

$\Rightarrow \underline{0.0098} \rightarrow \underline{0.98\%}$

It is good news that this is a rare disease because the disease is rare, even if the test is positive, its result is likely to be false positive. The chances that I actually have the disease is $\underline{0.98\%}$